CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Code 1: Draw with a sequence

Goal: I can put code blocks in order to show pictures in the order I planned.

35 minutes | 7 steps | Computer Science and Programming, Digital Literacy | Playbook B meeting 26

Ask | Plan | Build | Test | Improve

You are done when

Tell a partner what your pictures show, in order.

You will use

- Cart laptops, charged
- MakeCode for micro:bit in the browser, simulator only, if approved
- 5 by 5 grid cards and pencils for the no-device path

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Open MakeCode — 4 min

- Get your laptop.
- Open the MakeCode link on the board.

2 Step 2: Tell your partner — 1 min

- Tell a partner a set of steps you follow every day.

3 Step 3: Watch the blocks — 4 min

- Watch blocks snap together in order.
- Watch the simulator run them.

4 Step 4: Draw with code — 14 min

- Use show leds to make a picture.
- Add a second picture after it.
- Run it in the simulator.

5 Step 5: Check the order — 4 min

- Does it show in the order you planned?
- Swap two blocks and watch what changes.

6 Step 6: Save and shut — 5 min

- Name your project with your initials.
- Close the lid with two hands.
- Put the laptop back and plug it in.

7 Step 7: Show your result — 3 min

- Tell a partner what your pictures show, in order.

What to turn in

A program that shows your pictures in the order you planned.

No computer? Another way to do it

Write the steps on 5 by 5 grid cards and have a partner run them.

Finished early? Next challenge

Add a third picture and a pause between each.

Stay safe

Use only the MakeCode link the teacher gives. No sign-in is needed for the simulator.

Grade targets

Grade 5: Two pictures in order.
Grade 6: Three pictures with a pause, and predict the result before running.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.